

$2N \times 2N$ Block Transition Matrix M_{2N}

Up Phase ($i = 1 \dots N$)

Down Phase ($i = N+1 \dots 2N$)

Up

$$(1 - \beta)P$$

Up→Up
(stay driving)

$$\beta I$$

Up→Down
(park / slow)

Down

$$0$$

Down→Up
(impossible)

$$P$$

Down→Down
(local flow)